

Linear independence of distinct fair CDFs and a concatenation barrier

Research proof; independent mathematical reviews completed

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1 The general CDF theorem

The following extension of the rank-two calculation was found by the root collaborator and independently checked by the size-bounds agent.

Theorem 1. *Let $n \geq 3$ and let independent variables $X_1, \dots, X_n$ have continuous distribution functions $F_1, \dots, F_n$ on a common compact interval, rescaled to $[0, 1]$. Suppose every ordering of every triple has probability $1/6$. Then:*

- 1. Every nonzero linear relation among the F_i is supported on indices having the same distribution function.*
- 2. The distinct functions among $F_1, \dots, F_n$ are linearly independent.*
- 3. At most one distinct distribution function occurs more than once.*

Proof. For distinct j, k set

$$d_{kj} = \int_0^1 F_j^2 dF_k - \frac{1}{3}.$$

All Stieltjes integrals below are valid for continuous distribution functions. Triple fairness gives $\int F_i F_j dF_k = 1/3$ when i, j, k are pairwise distinct. Integration by parts also gives

$$\int F_k F_j dF_k = \frac{1}{2} - \frac{1}{2} \int F_k^2 dF_j = \frac{1}{3} - \frac{1}{2} d_{jk}.$$

Suppose $\sum_i a_i F_i = 0$. Evaluating at 1 gives $\sum_i a_i = 0$. Multiplying by F_j and integrating against dF_k , with $j \neq k$, therefore yields

$$0 = a_j d_{kj} - \frac{1}{2} a_k d_{jk}.$$

Swapping j, k gives $0 = a_k d_{jk} - a_j d_{kj}/2$. Consequently

$$a_j d_{kj} = a_k d_{jk} = 0 \quad (j \neq k). \quad (1)$$

For any two continuous distribution functions F, G , integration by parts yields the nonnegative identity

$$\int F^2 dG + \int G^2 dF - \frac{2}{3} = \frac{1}{2} \int (F - G)^2 d(F + G). \quad (2)$$

Its right-hand side is zero exactly when $F = G$. Indeed write $D = F - G$, $S = F + G$; monotonicity implies $|D(t) - D(s)| \leq S(t) - S(s)$ for $s \leq t$. If, say, $D(t) = d > 0$, choose the last $s < t$ with $D(s) = d/2$. Then $D \geq d/2$ on $[s, t]$ and $S(t) - S(s) \geq d/2$, so the integral on the right is at least $d^3/16 > 0$. The case $d < 0$ is analogous.

If a_j, a_k are both nonzero, (1) implies $d_{kj} = d_{jk} = 0$. Equation (2) then forces $F_j = F_k$. This proves the first assertion and, by choosing one representative of each distinct function, the second. Finally, if two different functions each occurred at least twice, the relation between two copies of the first function would imply $d_{kj} = 0$ in one direction, and the relation between two copies of the second would imply $d_{jk} = 0$. The same identity would force the two functions to coincide, a contradiction. $\square$

Corollary 2 (Exact rank of the block-count matrix). *Let a triple-fair word on $n \geq 3$ letters be a concatenation of permutation-fair blocks with positive multiplicities $v_i^{(g)}$. Put $C_i = \sum_g v_i^{(g)}$ and let Q be the matrix with columns*

$$q_i = (v_i^{(g)} / C_i)_{g=1}^s.$$

If the block-count matrix has rank r , then Q has exactly r distinct columns, those columns are linearly independent, and at most one of them is repeated. In particular, when all normalized letter profiles are distinct, the block-count vectors span all of $\mathbb{R}^n$.

Proof. Replace each block by a time interval of any fixed positive length, and each die by constant density in that interval with mass $v_i^{(g)} / C_i$. Independent choices of blocks have exactly the same probabilities as in the word. Conditional on choosing a common block, the order of every chosen subset is uniform by the fairness of the block, matching independent uniform positions in an interval. Thus all order probabilities, in particular triple fairness, are preserved. Linear relations between the resulting CDFs are exactly linear relations between the columns of Q : differentiation gives

one direction, and integration with zero initial values gives the other. Theorem 1 proves the result, since column normalization does not change the rank of the original count matrix. $\square$

2 The rank-two calculation and equalization consequences

The following direct proof is retained because it makes the rank-two obstruction explicit in two scalar integrals.

Theorem 3. *Let $n \geq 3$. Suppose independent random variables $X_1, \dots, X_n$ on $[0, 1]$ have bounded densities*

$$f_i(t) = 1 + a_i h(t), \quad \int_0^1 h(t) dt = 0,$$

where h is nonzero on a set of positive measure. If every ordering of every triple has probability $1/6$, then all but at most one of the coefficients a_i are equal.

Proof. Put $H(t) = \int_0^t h(x) dx$, so the distribution function of X_i is $F_i(t) = t + a_i H(t)$. Let

$$U = \int_0^1 t H(t) dt, \quad V = \int_0^1 H(t)^2 dt > 0.$$

The strict positivity of V follows from absolute continuity of H and the hypothesis on h . For pairwise distinct i, j, k , triple fairness implies

$$\Pr(X_i < X_j, X_k < X_j) = \int_0^1 F_i(t) F_k(t) f_j(t) dt = \frac{1}{3}.$$

Since $H(0) = H(1) = 0$, integration by parts gives

$$\int t^2 h = -2U, \quad \int t H h = -\frac{V}{2}, \quad \int H^2 h = 0.$$

Expanding the preceding probability therefore yields

$$2U(a_i + a_k - 2a_j) + V(2a_i a_k - a_j(a_i + a_k)) = 0. \quad (3)$$

Set $b_i = a_i + 2U/V$. Equation (3) is exactly

$$2b_i b_k = b_j(b_i + b_k).$$

Subtract the equation with middle index i from the one with middle index j to obtain

$$3b_k(b_i - b_j) = 0 \quad (i, j, k \text{ pairwise distinct}).$$

If all b_i are equal, the conclusion follows. Otherwise choose $b_i \neq b_j$. Every b_k outside this pair is zero. Since $n \geq 3$, there is such a k ; the equation with middle index j then gives $b_i b_j = 0$. Hence at most one b_i is nonzero, again proving the claim. $\square$

Theorem 4. *Let W be a concatenation of permutation-fair words $W_1, \dots, W_s$ on the same $n \geq 3$ letters, each having positive multiplicities. Write $v^{(g)} \in \mathbb{R}_{\geq 0}^n$ for the multiplicity vector of block g , and $C_i = \sum_g v_i^{(g)}$ for the total multiplicities. Suppose the vectors $v^{(g)}$ span a space of dimension at most two. If W is triple-fair, then there is an index e such that*

$$\frac{v_i^{(g)}}{C_i} = \frac{v_j^{(g)}}{C_j} \quad \text{for every } g \text{ and all } i, j \neq e.$$

Thus all but at most one letter must have the same distribution over the blocks.

Proof. The normalized block-mass vectors $q^{(g)} = (v_i^{(g)}/C_i)_{i=1}^n$ span a space of dimension at most two, contain the vector $\mathbf{1} = \sum_g q^{(g)}$ in their span, and have strictly positive entries. Choose any positive cell lengths λ_g summing to one and represent each die by constant density $q_i^{(g)}/\lambda_g$ on cell g . This model has exactly the same ordering probabilities as the original concatenation: the independent block choices have the same masses, and within each chosen block every restriction of its fair word has uniform relative order, just as independent uniforms in a common interval do.

One may choose cell lengths so that the density vectors lie on an affine line through $\mathbf{1}$. Indeed choose any linear functional L on the span with $L(\mathbf{1}) = 1$ and $L(q^{(g)}) > 0$ for all g , for example $L(x) = n^{-1} \sum_i x_i$. Put $\lambda_g = L(q^{(g)})$. Then each density vector $q^{(g)}/\lambda_g$ belongs to the affine slice $L(x) = 1$. If the span has dimension one, every density is $\mathbf{1}$ and the claim is immediate. Otherwise the slice is a line, so

$$f_i(t) = 1 + a_i h(t)$$

for a fixed vector a and a bounded piecewise-constant scalar h . The vector a is nonconstant because $L(a) = 0$ and $a \neq 0$; the density normalizations imply

$\int h = 0$. If h vanishes identically the claim is again immediate. Otherwise Theorem 3 says all but one a_i are equal, hence all but one density profile and block-mass profile are identical. $\square$

Corollary 5 (Two block types). *Let $c, d \in \mathbb{R}_{>0}^n$ and let a word be formed from positive scaled permutation-fair blocks of count vector c or d , with at least one block of each type. If the concatenation is triple-fair, then the ratios c_i/d_i are equal for all but at most one index i .*

Proof. Apply Theorem 4. On all nonexceptional indices i, j , both $c_i/C_i = c_j/C_j$ and $d_i/C_i = d_j/C_j$ hold, giving the assertion. $\square$

Corollary 6 (A transposition cannot equalize). *Let W be permutation-fair on $n \geq 3$ letters, and suppose letters i, j have different multiplicities. No concatenation using positive scaled copies of W and of its relabelling by the transposition (ij) , with at least one of each type, is triple-fair.*

Proof. The ratios of the two count vectors are $q, q^{-1}, 1$ on i, j and any other letter, where $q > 0$ and $q \neq 1$. These three ratios are distinct, contradicting Corollary 5. $\square$

Scope. The CDF theorem gives a structural rank obstruction for all such concatenations; its two-type corollary excludes simple transposition averaging. It gives no quantitative size lower bound for general equalization and no lower bound on the true equal-composition optimum. The same argument applies to triple-fair blocks; full permutation fairness of each block is more than is needed.